Hand-visible analytic derivations for 103 finite-ledger predicates

1 Scope and logical convention

This note exposes, in hand-checkable form, the predicates not assigned to the other five ledger packets. They are precisely

$$11 \text{ threshold predicates} + 48 \text{ zero predicates} + 43 \text{ lower-closure predicates} + 1 \text{ cap–74 predicate} = 103.$$

The omitted lower-closure predicate is the single six-cap tuple identity; it belongs to the cap-inventory packet. Every inequality below is proved from a displayed analytic inequality, integer arithmetic, rational cross multiplication, or positive-side squaring. An executable ledger may check the same arithmetic, but no executable output is used as a premise.

We use

$$\omega_0 = \frac{\sqrt{3}}{2\pi} - \frac{1}{6}, \quad C_0 = 1 + \frac{8\sqrt{2}}{15}, \quad \rho_* = \frac{\omega_0}{2C_0}, \quad \rho_0 = \frac{1}{\sqrt{337}}, \quad \sigma = \frac{3\omega_0}{4}, \quad \rho_c = \frac{1}{1+2\pi}, \quad d = \frac{\sqrt{337}}{2}.$$

The elementary enclosure

$$\frac{333}{106} < \pi < \frac{355}{113} < \frac{22}{7} \tag{1}$$

is analytic: the lower and middle bounds follow from alternating truncations of Machin's identity $\pi = 16\arctan(1/5) - 4\arctan(1/239)$, and the last one is immediate by cross multiplication. For reference, $355/113$ exceeds the upper Machin truncation by the positive rational numerator 45167474711 over denominator 189820334689453125. The classical integral

$$0 < \int_0^1 \frac{x^4(1-x)^4}{1+x^2}\,dx = \frac{22}{7} - \pi$$

also gives the final upper bound directly.

2 The eleven threshold predicates

First we obtain a rational two-sided enclosure for ω_0. The exact integer residuals

$$3\,1101^2 113^2 - 607^2 355^2 = 1998482 > 0, \qquad 25\,333^2 - 243\,106^2 = 41877 > 0$$

and (1) imply

$$\sqrt{3} > \frac{607\,355}{1101\,113} > \frac{607\pi}{1101}, \qquad 9\sqrt{3} < 5\pi.$$

Consequently

$$\frac{40}{367} < \omega_0 < \frac{1}{9}. \tag{2}$$

Indeed, $607/2202 - 1/6 = 40/367$, while $5/18 - 1/6 = 1/9$.

Since $2\,32^2 - 45^2 = 23 > 0$, one has $\sqrt{2} > 45/32$, hence $C_0 > 7/4$ and $\rho_* < 2/63$. The following chain consists entirely of strict rational comparisons, with the indicated radical comparisons justified by squaring:

$$\rho_* < \frac{2}{63} < \frac{1}{\sqrt{337}} < \frac{1}{18} < \frac{3}{40} < \frac{3\omega_0}{4} < \frac{1}{12} < \frac{7}{51} < \frac{1}{1+2\pi} < \frac{1}{7} < \frac{39}{50} < \frac{7}{8}.$$

Here

$$4 \cdot 337 < 63^2, qquad 18^2 < 337, qquad 367 < 400, \qquad 51 < 84, qquad 367 < 420,$$

and $7/51 < \rho_c < 1/7$ follows respectively from $\pi < 22/7$ and $\pi > 3$. This proves the five ratio-order rows. It also proves

$$\omega_0 < \frac{1}{9} < \frac{7}{51} < \rho_c.$$

Moreover

$$\omega_0 d > \frac{20\sqrt{337}}{367} > 1, \qquad 400 \cdot 337 - 367^2 = 111 > 0.$$

The four terminal comparisons follow from the same bounds:

$$\sqrt{114} < 18 < \frac{2}{\omega_0}, \qquad \frac{2}{\omega_0} < \frac{367}{20},$$

$$\rho_c < \frac{1}{7} < \frac{60}{367} < \frac{3\omega_0}{2}, \qquad \omega_0\sqrt{114} > \frac{40\sqrt{114}}{367} > 1.$$

The last positive-side squaring has residual

$$1600 \cdot 114 - 367^2 = 47711 > 0.$$

Thus the one-to-one threshold registry is

ID	source row	proved predicate
T1–T5	1–5	$\rho_* < \rho_0 < \sigma < \rho_c < 39/50 < 7/8$, one adjacent pair per row
T6	6	$\omega_0 < \rho_c$
T7	7	$\omega_0 d > 1$
T8	31	$\sqrt{114} < 2/\omega_0$
T9	32	$2/\omega_0 < 367/20$
T10	33	$\rho_c < 3\omega_0/2$
T11	34	$\omega_0\sqrt{114} > 1$

3 The forty-eight zero specializations

3.1 A recurrence and elementary tangent bounds

Put

$$\mathrm{j}_\ell(x) = \sqrt{\frac{\pi}{2x}} J_{\ell+1/2}(x), \qquad P_\ell(x) = x^{\ell+1}\mathrm{j}_\ell(x).$$

Then

$$P_{\ell+1} = (2\ell+1)P_\ell - x^2 P_{\ell-1}, \qquad P_0 = \sin x, quad P_1 = \sin x - x\cos x. \tag{3}$$

Write $P_\ell = A_\ell(t)\sin x + xC_\ell(t)\cos x$, $t = x^2$. The recurrence gives

$$\begin{aligned}
A_2 &= 3 - t, & C_2 &= -3,\\
A_3 &= 15 - 6t, & C_3 &= t - 15,\\
A_5 &= 945 - 420t + 15t^2, & C_5 &= -945 + 105t - t^2,\\
A_6 &= 10395 - 4725t + 210t^2 - t^3, & C_6 &= -10395 + 1260t - 21t^2,\\
A_7 &= 135135 - 62370t + 3150t^2 - 28t^3, & C_7 &= -135135 + 17325t - 378t^2 + t^3.
\end{aligned}$$

For $0 < s < \pi/2$, differentiation and integration give

$$\tan s > s, qquad \tan s > s + \frac{s^3}{3}. \tag{4}$$

For $0 < s < 1$, $\sin s < s$ and $\cos s > 1 - s^2/2$ give

$$\tan s < \frac{s}{1 - s^2/2}. \tag{5}$$

3.2 Sixteen half-period locations

For each internal face, the first displayed margin below is $w-(h_L/2)(22/7)$; the second is $(h_R/2)(333/106)-w$. Positivity therefore proves both strict half-period location rows without a decimal approximation.

(ℓ, n)	w	(h_L, h_R)	left margin	right margin
$(1,2)$	$77/10$	$(4,5)$	$99/70$	$163/1060$
$(2,2)$	9	$(5,6)$	$8/7$	$45/106$
$(3,2)$	$103/10$	$(6,7)$	$61/70$	$737/1060$
$(5,2)$	$129/10$	$(8,9)$	$23/70$	$1311/1060$
$(1,3)$	$21/2$	$(6,7)$	$15/14$	$105/212$
$(2,3)$	$61/5$	$(7,8)$	$6/5$	$97/265$
$(6,1)$	10	$(6,7)$	$4/7$	$211/212$
$(7,1)$	$23/2$	$(7,8)$	$1/2$	$113/106$

3.3 Sixteen half-period signs

At an integral or half-integral multiple of π, exactly one of $\sin x, \cos x$ vanishes. Direct substitution in the displayed polynomials gives the following complete sign table.

(ℓ, n)	h_L	left sign	(ℓ, n)	h_R	right sign
$(1,2)$	4	$-$	$(1,2)$	5	$+$
$(2,2)$	5	$-$	$(2,2)$	6	$+$
$(3,2)$	6	$-$	$(3,2)$	7	$+$
$(5,2)$	8	$-$	$(5,2)$	9	$+$
$(1,3)$	6	$+$	$(1,3)$	7	$-$
$(2,3)$	7	$+$	$(2,3)$	8	$-$
$(6,1)$	6	$+$	$(6,1)$	7	$-$
$(7,1)$	7	$+$	$(7,1)$	8	$-$

Here are explicit polynomial witnesses for the only signs not immediate from $A_1 = 1, C_1 = -1, A_2 = 3 - t, C_2 = -3, A_3 = 15 - 6t, C_3 = t - 15$. From (1), at $x = 4\pi$ one has $144 < t < 159$, at $x = 9\pi/2$ one has $t > 182$, and at $x = 7\pi/2$ one has $110 < t < 121$. Monotonicity on these rational intervals and direct evaluation give

$$C_5(t) < C_5(144) = -6561, qquad A_5(t) > A_5(182) = 421365,$$

$$C_6(t) < C_6(81) = -46116, qquad A_6(t) > A_6(110) = 700645,$$

$$A_7(t) < A_7(110) = -5878565, qquad C_7(t) < C_7(144) = -2492559.$$

For example, $A_6' = -4725 + 420t - 3t^2$ is positive on $[110, 121]$, while A_7' is negative there; the other asserted monotonicities follow at once from their displayed derivatives. Combining these signs with the quadrant signs of sine and cosine proves all sixteen entries.

3.4 Eight rational-wall signs

Substitution in (3) reduces each rational-wall sign to one of the following exact tangent comparisons. The shifted angle is always in $(0, \pi/2)$, as already proved by the location table.

(ℓ, n)	wall	hand certificate	$\operatorname{sgn} \mathrm{j}_\ell(w)$
$(1, 2)$	$77/10$	For $u = 5\pi/2 - w$, $\tan u > 3/20 > 10/77 = 1/w$; hence $\tan(w - 2\pi) = \cot u < w$.	$-$
$(2, 2)$	9	For $u = 3\pi - w$, $\tan u > 21/50 > 9/26 = 3w/(w^2 - 3)$.	$-$
$(3, 2)$	$103/10$	For $u = 7\pi/2 - w$, $\tan u > 69/100 > 621540/938227 = (6w^2 - 15)/(w(w^2 - 15))$.	$-$
$(5, 2)$	$129/10$	For $y = w - 4\pi$, $0 < y < 17/50$, so $\tan y < 1700/4711 < 177800829/427700150 = -wC_5/A_5$.	$-$
$(1, 3)$	$21/2$	For $u = 7\pi/2 - w$, $\tan u > 49/100 > 2/21 = 1/w$, hence $\tan(w - 3\pi) < w$.	$+$
$(2, 3)$	$61/5$	For $u = 4\pi - w$, $\tan u > 9/25 > 915/3646 = 3w/(w^2 - 3)$.	$+$
$(6, 1)$	10	For $y = w - 3\pi$, $4/7 < y < 29/50$, so $\tan y < 2900/4159 < 188790/127579 = -wC_6/A_6$.	$+$
$(7, 1)$	$23/2$	For $u = 4\pi - w$, $\tan u > u + u^3/3 > 546377/375000 > 3153151489/2276518664 = wC_7/A_7$.	$+$

For complete cross-multiplication visibility, the four less immediate positive gaps in that table are

$$\frac{69}{100} - \frac{621540}{938227} = \frac{2583663}{93822700},$$
$$\frac{177800829}{427700150} - \frac{1700}{4711} = \frac{110529450419}{2014895406650},$$
$$\frac{188790}{127579} - \frac{2900}{4159} = \frac{415198510}{530601061},$$
$$\frac{546377}{375000} - \frac{3153151489}{2276518664} = \frac{3837851856583}{53355906187500}.$$

The four small gaps are respectively $31/1540, 24/325, 829/2100, 9939/91150$.

3.5 Six Lorch rows and two wall-order rows

For $\nu = \ell + 1/2$, Lorch's strict analytic first-zero estimate is

$$j_{\nu,1}^2 > \frac{24(\nu+1)^2}{1 - 2\nu + \sqrt{(2\nu+3)(2\nu+11)}} - 2(\nu^2 - 1) =: R_\ell.$$

After rationalization,

$$R_\ell = \frac{3(2\ell+3)\sqrt{(\ell+2)(\ell+6)} - 2\ell^2 + \ell + 6}{4}.$$

All sides being squared are positive, and the following integer residuals prove the three required strict specializations:

ℓ	wall c	positive squared residual proving $R_\ell > c^2$
8	71/6	$1026^2 \cdot 35 - 6067^2 = 35171$
9	64/5	$1575^2 \cdot 165 - 20059^2 = 6939644$
10	69/5	$3450^2 \cdot 3 - 5911^2 = 767579$

Since $\mathsf{j}_\ell(x) > 0$ for sufficiently small $x > 0$, and its first positive zero is strictly to the right of c, these same three rows imply $\mathsf{j}_\ell(c) > 0$. Thus the three executable sign rows are analytic corollaries, not independent numerical assumptions. Finally,

$$\frac{103}{10} - \frac{81}{8} = \frac{7}{40} > 0, \qquad \frac{61}{5} - \frac{21}{2} = \frac{17}{10} > 0.$$

The one-to-one zero registry is as follows. Number the eight internal faces in the order of the location table. For face r, IDs $Z_{5r-4}, \ldots, Z_{5r}$ are, in order: right of the left half-period, left of the right half-period, left half-period sign, right half-period sign, and rational-wall sign. This bijects IDs $Z1$–$Z40$ with the first forty source rows. The remaining bijection is

ID	source row	predicate
Z41	41	Lorch specialization $\ell = 8, c = 71/6$
Z42	42	positive sign at $\ell = 8, c = 71/6$
Z43	43	Lorch specialization $\ell = 9, c = 64/5$
Z44	44	positive sign at $\ell = 9, c = 64/5$
Z45	45	Lorch specialization $\ell = 10, c = 69/5$
Z46	46	positive sign at $\ell = 10, c = 69/5$
Z47	47	$103/10 > 81/8$
Z48	48	$61/5 > 21/2$

4 The forty-three lower-closure predicates

4.1 Eight activation and exclusion rows

The complete arithmetic is

$$\begin{aligned} (23/2)^2 - 114 &= 73/4 > 0, & 10^2 &= 100, \\ (129/10)^2 - 114 &= 5241/100 > 0, & 103/10 - 81/8 &= 7/40 > 0, \\ (81/8)^2 + 8 &= 7073/64, & (81/8)^2 + 18 - 114 &= 417/64 > 0, \\ (21/2)^2 + 4 - 114 &= 1/4 > 0, & 16\pi^2 &> 16 \cdot 3^2 > 114. \end{aligned}$$

These are exactly lower IDs L1–L8, corresponding to source rows 1–8.

4.2 Five Weyl payments

For $0 < \rho < \rho_c < 1/7$, (1) yields the uniform analytic lower bound

$$W(\rho, K) = \frac{2}{9\pi}(1 - \rho^3)K^3 > \frac{38}{539}K^3. \tag{6}$$

Using $d > 9$ and $\sqrt{7073}/8 > 21/2$, the five payments are the following exact positive reserves:

ID	source row	threshold/cap	reserve from (6)
L9	10	$K > d,\ 46$	$(38/539)9^3 - 46 = 2908/539$
L10	11	$K > 10,\ 59$	$(38/539)10^3 - 59 = 6199/539$
L11	12	$K > 81/8,\ 66$	$(38/539)(81/8)^3 - 66 = 990435/137984$
L12	13	$K > 21/2,\ 69$	$(38/539)(21/2)^3 - 69 = 555/44$
L13	14	$K > \sqrt{7073}/8,\ 78$	$(38/539)(21/2)^3 - 78 = 159/44$

4.3 The cap–395 cell: eighteen rows

Put $R = 367/20$, $z_\ell = \ell + 1/2$. The analytic turning estimate

$$G_K(z) < \frac{(K-z)^{3/2}}{3\sqrt{K}}$$

is increasing in K, so on $K < R$ a strict cap c_ℓ follows from

$$\frac{(R-z_\ell)^3}{9R} = \frac{(357-20\ell)^3}{1321200} < \left(c_\ell + \frac{3}{4}\right)^2. \tag{7}$$

The following table gives all fifteen caps and the exact positive numerator $82575(4c_\ell+3)^2 - (357-20\ell)^3$ obtained by clearing the positive denominators in (7).

ℓ	0	1	2	3	4	5	6	7
c_ℓ	6	5	5	4	4	3	3	3
margin	14697882	5409422	11827162	3611502	8555642	1604782	5267322	8361062

ℓ	8	9	10	11	12	13	14
c_ℓ	2	2	1	1	1	1	0
margin	2346202	4446342	176282	1474822	2444562	3133502	286642

Thus L15–L29 correspond bijectively to source rows 16–30, with L$(15+\ell)$ the ℓ-th table column. Since $G_R(z)$ is strictly decreasing in z, the $c_{14} = 0$ row removes every $\ell \geq 14$ tail; this is L30/source row 31.

The weighted cap itself, L14/source row 15, is visible from the contributions

$$6, 15, 25, 28, 36, 33, 39, 45, 34, 38, 21, 23, 25, 27, \qquad \sum_{\ell=0}^{13}(2\ell+1)c_\ell = 395.$$

Finally, on the exceptional cell $\rho K \geq 5/2$, with $\rho < 11/80$ and $1/\pi > 113/355$,

$$\begin{aligned} W(\rho, K) &> \frac{2}{9}\frac{113}{355}\left(\frac{5}{2}\right)^3\left[\left(\frac{80}{11}\right)^3 - 1\right] \\ &= \frac{160293325}{378004} = 395 + \frac{10981745}{378004} > 395. \end{aligned}$$

This is L31/source row 32.

4.4 Twelve floor and boundary-owner rows

The nine finite arithmetic rows are instances of one identity:

$$\frac{3}{2}(p+1) \leq 2p + \frac{1}{2} \quad (p \geq 2), qquad \left(2p + \frac{1}{2}\right) - \frac{3}{2}(p+1) = \frac{p-2}{2} \geq 0. \tag{8}$$

At $p = 2$, $(3/2)3 = 9/2$; this is L32/source row 33. For $p = 2, \dots, 9$, formula (8) is respectively L33–L40/source rows 34–41.

The half-integer face is exact: $5/2 = 2 + 1/2$. Assigning it to the high side makes the interface remainder vanish and leaves the exceptional cell closed; this is L41/source row 42. At $K = 2/\omega_0$,

$$p = \lfloor \omega_0 K \rfloor = 2, \qquad \rho K \le \frac{2\rho_c}{\omega_0} < 3$$

by T10. This proves L42/source row 43. Moreover

$$\rho K - \frac{1}{2} < \frac{5}{2}, qquad m = \left\lfloor \rho K - \frac{1}{2} \right\rfloor \le 2 \le 2p - 1 = 3,$$

and the strict exceptional condition $K < 2/\omega_0$ fails at equality. Thus the equality face is aggregate-owned, proving L43/source row 44. More generally, if $K \ge 2/\omega_0$, then $p \ge 2$ and

$$\rho K < \frac{3}{2}(p+1) \le 2p + \frac{1}{2}, \qquad m \le 2p - 1,$$

so the finite instances conceal no unproved large-p case.

5 The supplemental cap–74 incompatibility

The propagated ball wall requires $z^2 > 2409/100$, while the product wall requires $z^2 < 70/3$. They are incompatible because

$$\frac{2409}{100} - \frac{70}{3} = \frac{7227 - 7000}{300} = \frac{227}{300} > 0.$$

This is supplemental ID S1 and is a hand proof, not merely a reported certificate value.

6 Census and source-visibility verdict

family	count	IDs	hand-visible status before this note
threshold remainder	11	T1–T11	printed, with a few implications compressed
zero specializations	48	Z1–Z48	tables printed; several recurrence steps compressed
lower remainder	43	L1–L43	payments printed; floor instances generated implicitly
supplemental cap–74	1	S1	printed exactly
total	103		

No predicate in this 103-row allocation is an unsupported executable-only mathematical premise. Before this note, the principal visibility gaps were the one-to-one expansion of the forty zero sign/location rows, the fact that the three Lorch-wall signs are logical corollaries of the strict first-zero bounds, and the expansion of the twelve floor/boundary rows. The displayed recurrences, rational margins, general floor identity, and exact ownership arguments close those presentation gaps.